

HSC Practice Questions

Annuities

- 1 Julie invests \$12 500 in a savings account. Interest is paid at a fixed monthly rate. At the end of each month, after the monthly interest is added, Julie makes a deposit of \$500.

Julie has created a spreadsheet to show the activity in her savings account. The details for the first 6 months are shown.

Month	Amount in account at beginning of month	Monthly interest	Deposit	Amount in account at end of month
1	12 500.00	18.75	500	13 018.75
2	13 018.75	19.53	500	13 538.28
3	13 538.28	20.31	500	14 058.59
4	14 058.59	21.09	500	14 579.68
5	14 579.68	21.87	500	15 101.55
6	15 101.55	22.65	500	15 624.20
7			500	

By finding the monthly rate of interest, complete the final row above for the 7th month. (3 marks)

(Q21, 2021 HSC) Medium

- 2 Present value interest factors for an annuity of \$1 for various interest rates (r) and numbers of periods (N) are given in the table.

Table of present value interest factors				
$N \backslash r$	Interest rate per period as a decimal			
	0.001	0.00125	0.0015	0.00175
300	259.07072	250.03980	241.43789	233.24180
330	280.95771	270.26900	260.13532	250.52386
360	302.19816	289.75411	278.01062	266.92278

A bank lends Martina \$500 000 to purchase a home, with interest charged at 1.5% per annum compounding monthly. She agrees to repay the loan by making equal monthly repayments over a 30-year period.

How much should the monthly payment be in order to pay off the loan in 30 years? Give your answer correct to the nearest cent. (2 marks)

(Q31, 2021 HSC) Medium

- 3 A table of future value interest factors for an annuity of \$1 is shown.

Number of periods	Interest rate per period				
	0.25%	0.5%	0.75%	1%	1.25%
2	2.0025	2.0050	2.0075	2.0100	2.0125
4	4.0150	4.0301	4.0452	4.0604	4.0756
6	6.0376	6.0755	6.1136	6.1520	6.1907
8	8.0704	8.1414	8.2132	8.2857	8.3589
10	10.1133	10.2280	10.3443	10.4622	10.5817

Simone deposits \$1000 into a savings account at the end of each year for 8 years. The

interest rate for these 8 years is 0.75% per annum, compounded annually.

After the 8th deposit, Simone stops making deposits but leaves the money in the savings account. The money in her savings account then earns interest at 1.25% per annum, compounded annually, for a further two years.

Find the amount of money in Simone's savings account at the end of ten years. (3 marks)

CO (Q40, 2021 HSC) Hard

- 4 An annuity consists of ten payments, each equal to \$1000. Each payment is made on 30 June each year from 2021 through to 2030 inclusive.

The rate of compound interest is 5% per annum.

The present value of the annuity is calculated at 30 June 2020.

The future value of the annuity is calculated at 30 June 2030.

Without performing any calculations, which of the following statements is true?

- A Present value of the annuity < \$10 000 < future value of the annuity
- B \$10 000 < present value of the annuity < future value of the annuity
- C Future value of the annuity < \$10 000 < present value of the annuity
- D \$10 000 < future value of the annuity < present value of the annuity (1 mark)

(Q14, 2020 HSC) Medium

- 5 Tina inherits \$60 000 and invests it in an account earning interest at a rate of 0.5% per month. Each month, immediately after the interest has been paid, Tina withdraws \$800.

The amount in the account immediately after the n th withdrawal can be determined using the recurrence relation

$$A_n = A_{n-1}(1.005) - 800,$$

where $n = 1, 2, 3, \dots$ and $A_0 = 60 000$.

- a Use the recurrence relation to find the amount of money in the account immediately after the third withdrawal. (2 marks) Medium
- b Calculate the amount of interest earned in the first three months. (2 marks) Hard

CO (Q34, 2020 HSC)

HSC Practice Questions

Annuities

- 6 Wilma deposited a lump sum into a new bank account which earns 2% per annum compound interest.

Present value interest factors for an annuity of \$1 for various interest rates (r) and numbers of periods (N) are given in the table.

Table of present value interest factors

$N \backslash r$	Interest rate per period as a decimal			
	0.01	0.015	0.02	0.025
10	9.471	9.222	8.983	8.752
20	18.046	17.169	16.351	15.589
30	25.808	24.016	22.396	20.930

Wilma was able to make the following withdrawals from this account.

- \$1000 at the end of each year for twenty years (starting one year after the account is opened)
- \$3000 each year for ten years starting 21 years after the account is opened.

Calculate the minimum lump sum Wilma must have deposited when she opened the new account. (3 marks)

(Q37, 2020 HSC) **Hard**

- 7 The table shows the future values of an annuity of \$1 for different interest rates for 4, 5 and 6 years. The contributions are made at the end of each year.

Future value of an annuity of \$1

Years	Interest rate per annum			
	1%	2%	3%	4%
4	4.060	4.122	4.184	4.246
5	5.101	5.204	5.309	5.416
6	6.152	6.308	6.468	6.633

An annuity account is opened and contributions of \$2000 are made at the end of each year for 7 years.

For the first 6 years, the interest rate is 4% per annum, compounding annually. For the 7th year, the interest rate increases to 5% per annum, compounding annually.

Calculate the amount in the account immediately after the 7th contribution is made. (3 marks)

(Q42, 2019 HSC) **Hard**

- 8 Sally is currently aged 50 and is planning to retire at the end of the year in which she turns 65. Her current self-managed super fund has a balance of \$530 000 and is delivering 3.2% compound interest. At the start of the year she makes a payment of \$8600 into the fund. However, Sally plans to increase this annual deposit by $n\%$ each year. A table for the annuity is shown below including the balance of the annuity at the end of each year.

Payment number	Payment received	Interest earned	Principal increase	Balance of annuity
0	\$0.00	\$0.00	\$0.00	\$530 000.00
1	\$8600.00	\$17 235.20	\$25 835.20	\$555 835.20
2	\$8772.00	\$18 067.43	\$26 839.43	\$582 674.63
3	\$8947.44	\$18 931.91	\$27 879.35	\$610 553.98
4	\$9126.39	\$19 829.77	\$28 956.16	\$639 510.14
5	\$9308.92	\$20 762.21	\$30 071.13	\$669 581.26
6	\$9495.09	\$21 730.44	\$31 225.54	\$700 806.80
7	\$9685.00	\$22 735.74	\$32 420.73	\$733 227.54
8	\$9878.70	\$23 779.40	\$33 658.10	\$766 885.63
9	\$10 076.27	\$24 862.78	\$34 939.05	\$801 824.68
10	\$10 277.80	\$25 987.28	\$36 265.08	\$838 089.76
11	\$10 483.35	\$27 154.34	\$37 637.69	\$875 727.45
12	\$10 693.02	\$28 365.46	\$39 058.47	\$914 785.93
13	\$10 906.88	A		B
14	\$11 125.02	\$30 926.08	\$42 051.10	\$997 366.07
15	\$11 347.52	\$32 278.83	\$43 626.35	\$1 040 992.42

- a Find the value of n .
 b Find **A** and **B**.
 c Sally has decided to delay retirement for one year.
 i What will be her contribution for payment 16?
 ii What will be the *new* balance for the annuity at the end of her 66th year?

(1 mark) **Easy**

(2 marks) **Medium**

(1 mark) **Medium**

(1 mark) **Medium**

Bonus question (see page iv)

HSC Practice Questions

Annuities

- 9 Mark needs \$8000 to go on a holiday in three years time. He has a holiday savings account with a balance of \$600. He arranges to deposit \$150 into this account at the end of each month for the next 3 years. He earns 6% per annum interest on the money in the account, compounded monthly. The table below shows future values of an annuity of \$1 and can be used in your calculations.

Period	Table of future value interest factors							
	Interest rate per period							
	0.50%	1.00%	1.50%	2.00%	3.00%	4.00%	5.00%	6.00%
5	5.0503	5.1010	5.1523	5.2040	5.3091	5.4163	5.5256	5.6371
6	6.0755	6.1520	6.2296	6.3081	6.4684	6.6330	6.8019	6.9753
7	7.1059	7.2135	7.3230	7.4343	7.6625	7.8983	8.1420	8.3938
8	8.1414	8.2857	8.4328	8.5830	8.8923	9.2142	9.5491	9.8975
12	12.3356	12.6825	13.0412	13.4121	14.1920	15.0258	15.9171	16.8699
24	25.4320	26.9735	28.6335	30.4219	34.4265	39.0826	44.5020	50.8156
36	39.3361	43.0769	47.2760	51.9944	63.2759	77.5983	95.8363	119.1209
48	54.0978	61.2226	69.5652	79.3535	104.4084	139.2632	188.0254	256.5645
60	69.7700	81.6697	96.2147	114.0515	163.0534	237.9907	353.5837	533.1282
72	86.4089	104.7099	128.0772	158.0570	246.6672	396.0566	650.9027	1089.6286

Will Mark have enough money for his trip at the end of the 3 years? Justify your answer with calculations.

(3 marks)

Bonus question **Hard**

- 10 The table shows the present value interest factors for various interest rates and periods of time.

Period	Table of present value interest factors						
	Interest rate per period						
	0.50%	1.00%	1.25%	1.50%	2.00%	3.00%	4.00%
20	18.9874	18.0456	17.5993	17.1686	16.3514	14.8775	13.5903
30	27.7941	25.8077	24.8889	24.0158	22.3965	19.6004	17.2920
40	36.1722	32.8347	31.3269	29.9158	27.3555	23.1148	19.7928
50	44.1428	39.1961	37.0129	34.9997	31.4236	25.7298	21.4822
120	90.0735	69.7005	61.9828	55.4985	45.3554	32.3730	24.7741
180	118.5035	83.3217	71.4496	62.0956	48.5844	33.1703	24.9785
240	139.5808	90.8194	75.9423	64.7957	49.5686	33.3057	24.9980

Mel is 40 years old today and plans to retire when she turns 60 years old. In retirement, she would like to receive an annual annuity of \$60000 until she turns 90. What single amount should she deposit today in a savings account in order to have an investment large enough to receive her annual annuity in retirement, assuming an interest rate of 4% per annum for the entire 50 years? Justify your answers with calculations.

(4 marks)

Bonus question **Hard**

- 11 Tracey wants an annuity of \$15000 every 3 months for the next 30 years. Her financial institution can guarantee an interest rate of 2% p.a. compounded quarterly for the life of the annuity. Use the table below to find the amount Tracey will need to invest today to provide the annuity.

Period	Table of present value interest factors						
	Interest rate per period						
	0.50%	1.00%	1.25%	1.50%	2.00%	3.00%	4.00%
20	18.9874	18.0456	17.5993	17.1686	16.3514	14.8775	13.5903
30	27.7941	25.8077	24.8889	24.0158	22.3965	19.6004	17.2920
40	36.1722	32.8347	31.3269	29.9158	27.3555	23.1148	19.7928
50	44.1428	39.1961	37.0129	34.9997	31.4236	25.7298	21.4822
120	90.0735	69.7005	61.9828	55.4985	45.3554	32.3730	24.7741
180	118.5035	83.3217	71.4496	62.0956	48.5844	33.1703	24.9785
240	139.5808	90.8194	75.9423	64.7957	49.5686	33.3057	24.9980

(2 marks)

Bonus question **Medium**

HSC Practice Questions

Annuities

- 12 A table of future value interest factors for an annuity of \$1 is shown.

Period	Table of future value interest factors				
	Interest rate per period				
	1%	2%	3%	4%	5%
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019

An annuity involves contributions of \$12 000 per annum for 5 years. The interest rate is 4% per annum, compounded annually.

- i Calculate the future value of this annuity. (1 mark) **Easy**
- ii Calculate the interest earned on this annuity. (1 mark) **Hard**

(Q27c, 2017 HSC)

- 13 The table gives the contribution per period for an annuity with a future value of \$1 at different interest rates and different periods of time.

Contribution per period for an annuity with a future value of \$1

Number of periods	Interest rate (% per period)					
	0.25%	0.5%	0.75%	1%	1.25%	1.5%
6	0.1656	0.1646	0.1636	0.1625	0.1615	0.1605
12	0.0822	0.0811	0.0800	0.0788	0.0778	0.0767
18	0.0544	0.0532	0.0521	0.0510	0.0499	0.0488
24	0.0405	0.0393	0.0382	0.0371	0.0360	0.0349
30	0.0321	0.0310	0.0298	0.0287	0.0277	0.0266
36	0.0266	0.0254	0.0243	0.0232	0.0222	0.0212

Margaret needs to save \$75 000 over 6 years for a deposit on a new apartment. She makes regular quarterly contributions into an investment account which pays interest at 3% pa.

How much will Margaret need to contribute each quarter to reach her savings goal? (2 marks)

(Q28d, 2016 HSC) **Medium**

- 14 The table gives the present value interest factors for an annuity of \$1 per period, for various interest rates (r) and numbers of periods (N).

$r \backslash N$	Table of present value interest factors				
	Interest rate per period (as a decimal)				
	0.0075	0.0080	0.0085	0.0090	0.0095
70	54.30462	53.43960	52.59397	51.76724	50.95891
71	54.89293	54.00754	53.14226	52.29657	51.46995
72	55.47685	54.57097	53.68593	52.82118	51.97618
73	56.05643	55.12993	54.22502	53.34111	52.47764
74	56.63169	55.68446	54.75957	53.85641	52.97438

- i Oscar plans to invest \$200 each month for 74 months. His investment will earn interest at the rate of 0.0080 (as a decimal) per month. Use the information in the

table to calculate the present value of this annuity. (1 mark) **Medium**

- ii Lucy is using the same table to calculate the loan repayment for her car loan. Her loan is \$21 500 and will be repaid in equal monthly repayments over 6 years. The interest rate on her loan is 10.8% per annum. Calculate the amount of each monthly repayment, correct to the nearest dollar. (2 marks) **Hard**

(Q30c, 2015 HSC)

- 15 A table of future value interest factors is shown.

Period	Table of future value interest factors				
	Interest rate per period				
	1%	2%	3%	4%	5%
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101

A certain annuity involves making equal contributions of \$25 000 into an account every 6 months for 2 years at an interest rate of 4% per annum.

Based on the information provided, what is the future value of this annuity?

- A \$50 500 B \$51 000
C \$103 040 D \$106 162 (1 mark)

(Q21, 2014 HSC) **Medium**

- 16 Zina opened an account to save for a new car. Six months after opening the account, she made her first deposit of \$1200 and continued depositing \$1200 at the end of each six month period. Interest was paid at 3% per annum, compounded half-yearly.

How much was in Zina's account two years after first opening it?

- A \$4909.08 B \$4982.72
C \$5018.16 D \$5094.55 (1 mark)

(Q23, 2013 HSC) **Medium**

- 17 An amount of \$5000 is invested at 10% per annum, compounded six-monthly.

Period	Compounded values of \$1			
	Interest rate per period			
	1%	5%	10%	20%
1	1.010	1.050	1.100	1.150
2	1.020	1.103	1.210	1.323
3	1.030	1.158	1.331	1.521
4	1.041	1.216	1.464	1.750
5	1.051	1.276	1.611	2.011
6	1.062	1.340	1.772	2.313

Use the table to find the value of this investment at the end of three years. (2 marks)

(Q23c, 2011 HSC) **Hard**

HSC Practice Questions

Annuities

- 18 The table shows the future value of a \$1 annuity at different interest rates over different numbers of time periods.

Future values of a \$1 annuity

Time Period	Interest rate				
	1%	2%	3%	4%	5%
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

- What would be the future value of a \$5000 per year annuity at 3% per annum for 6 years, with interest compounding yearly? (1 mark) **Easy**
- What is the value of an annuity that would provide a future value of \$407 100 after 7 years at 5% per annum compound interest? (1 mark) **Medium**
- An annuity of \$1000 per quarter is invested at 4% per annum, compounded quarterly for 2 years. What will be the amount of interest earned? (3 marks) **Hard**

(Q27a, 2009 HSC)

- 19 Rod is saving for a holiday. He deposits \$3600 into an account at the end of every year for four years. The account pays 5% per annum interest, compounding annually.

The table shows future values of an annuity of \$1.

Future values of an annuity of \$1

End of year	Interest rate				
	1%	2%	3%	4%	5%
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

- Use the table to find the value of Rod's investment at the end of four years. (2 marks) **Easy**
- How much interest does Rod earn on his investment over the four years? (2 marks) **Medium**

(Q26b, 2005 HSC)

Annuities

- 1 Monthly interest rate

$$= \frac{18.75}{12\,500.00} \times 100\%$$

$$= 0.15\%$$

The amount at the beginning of month 7 is 15624.20.

$$\text{Interest} = 0.0015 \times 15\,624.20$$

$$= 23.4363$$

$$= 23.44 \quad (2 \text{ d.p.})$$

Amount at the end of the month

$$= 15\,624.20 + 23.44 + 500$$

$$= 16\,147.64$$

7	15624.20	23.44	500	16147.64
---	----------	-------	-----	----------

(3 marks)

- 2 30 years = 30 × 12 months

$$= 360 \text{ months}$$

So $N = 360$.

$$\text{Interest rate per period} = \frac{0.015}{12}$$

$$= 0.00125$$

Present value factor is 289.75411 ✓

Monthly payment

$$= \$500\,000 \div 289.75411$$

$$= \$1\,725.601\,06\dots$$

$$= \$1\,725.60 \quad (\text{nearest cent})$$

(2 marks)

- 3 Interest rate = 0.75%,
8 periods

Future value factor = 8.2132 ✓

After the 8th deposit,

$$FV = \$1\,000 \times 8.2132$$

$$= \$8\,213.20$$

For final 2 years, $PV = \$8\,213.20$,

$$r = 0.0125$$

$$FV = PV(1 + r)^n$$

$$= \$8\,213.20 \times (1 + 0.0125)^2$$

$$= \$8\,419.81 \quad (\text{nearest cent})$$

The amount of money in the account at the end of 10 years is \$8419.81. ✓

(3 marks)

- 4 Total payments = \$10000

The future value must be more than that as interest is earned.

The present value must be less than \$10000 as it is the single amount that could be invested at the same rate.

Answer A

- 5 a $A_n = A_{n-1}(1.005) - 800$

$$A_0 = 60\,000$$

$$A_1 = 60\,000(1.005) - 800$$

$$= 59\,500$$

$$A_2 = 59\,500(1.005) - 800$$

$$= 58\,997.5$$

$$A_3 = 58\,997.5(1.005) - 800$$

$$= 58\,492.4875$$

$$= 58\,492.49 \quad (2 \text{ d.p.})$$

The amount of money in the account immediately after the third withdrawal is \$58492.49. ✓

(2 marks)

- b Total amount withdrawn

$$= 3 \times \$800$$

$$= \$2\,400$$

Interest earned

$$= \$58\,492.49 + \$2\,400 - \$60\,000$$

$$= \$892.49$$

(2 marks)

- 6 \$1000 each year for 20 years:

$$r = 0.02, N = 20$$

Present value factor = 16.351

$$\text{Present value} = \$1\,000 \times 16.351$$

$$= \$16\,351$$

So Wilma will need to deposit \$16351 to withdraw \$1000 per year for the next 20 years.

\$3000 each year for 10 years

$$r = 0.02, N = 10$$

Present value factor = 8.983

$$\text{Present value} = \$3\,000 \times 8.983$$

$$= \$26\,949$$

So Wilma will need to have \$26949 in the account at the end of 20 years to be able to withdraw \$3000 each year for the following 10 years.

Amount required at the beginning of the 20 years

$$= \frac{\$26\,949}{(1 + 0.02)^{20}}$$

$$= \$18\,135.9044\dots$$

$$= \$18\,136 \quad (\text{nearest dollar})$$

Minimum deposit

$$= \$16\,351 + \$18\,136$$

$$= \$34\,487$$

(3 marks)

- 7 Future value at end of 6 years

$$= 6.633 \times \$2\,000$$

$$= \$13\,266$$

Interest for 7th year

$$= 0.05 \times \$13\,266$$

$$= \$663.30$$

Total at end of 7 years

$$= \$13\,266 + \$663.30 + \$2\,000$$

$$= \$15\,929.30$$

(3 marks)

- 8 a Difference = 8772 - 8600

$$= 172$$

$$\text{Percentage} = \frac{172}{8600} \times 100\%$$

$$= 2$$

$$\therefore n = 2\%$$

(1 mark)

- b $A: 0.032 \times (914\,785.93$

$$+ 10\,906.88)$$

$$= 29\,622.17$$

$$\therefore A = \$29\,622.17$$

$$B: 914\,785.93 + 10\,906.88$$

$$+ 29\,622.17$$

$$= 955\,314.98$$

$$\therefore B = \$955\,314.98$$

(2 marks)

- c i $11\,347.52 \times 1.02$

$$= 11\,574.47$$

$\therefore$ contribution of

$$\$11\,574.47$$

(1 mark)

- ii $(1\,040\,992.42 + 11\,574.47)$

$$\times 1.032$$

$$= 1\,086\,249.03$$

$\therefore$ new balance of

$$\$1\,086\,249.03.$$

(1 mark)

- 9 6% per annum

= 0.06 per annum

$$= \frac{0.06}{12} \text{ per month}$$

$$= 0.005$$

$$3 \text{ years} = 3 \times 12 \text{ months}$$

$$= 36 \text{ months}$$

Mark's account has two parts:

\$600 earning 6% p.a.

compounded monthly for

3 years:

$$FV = PV(1 + r)^n$$

$$= 600(1.005)^{36}$$

$$= 718.008\,3149\dots$$

$$\therefore \$718.01$$

Also, \$150 deposited each

month. Using the table ...

$$FV = 150 \times 39.336$$

$$= 5900.4$$

$$\therefore \$5900.40$$

$$\text{Total} = \$718.01 + \$5900.40$$

$$= \$6618.41$$

$\therefore$ Mark will not have enough money. ✓

(3 marks)

- 10 From 60 years old to 90 years old

is 30 years. Also, 4%. ✓

$$PV = 60\,000 \times 17.2920$$

$$= 1\,037\,520$$

$\therefore$ Mel needs \$1037520 in her

savings account when she turns

60. ✓

Using $FV = 1\,037\,520$, $r = 0.04$ and $n = 20$:

$$FV = PV(1 + r)^n$$

$$1\,037\,520 = PV(1.04)^{20}$$

$$PV = \frac{1\,037\,520}{1.04^{20}}$$

$$= 473\,510.5844\dots$$

$$= 473\,511$$

(nearest whole)

$\therefore$ Mel needs to deposit \$473511 in her savings account today. ✓

(4 marks)

Annuities

- 11 2% p.a. = 0.5% per quarter
 30 years = 120 quarters
 $PV = 15000 \times 90.0735$ ✓
 $= 1\,351\,102.50$
 $\therefore$ Tracey should invest
 $\$1\,351\,102.50$. ✓
 (2 marks)

- 12 i Period 5 years, interest rate
 = 4%
 Interest rate factor is 5.4163.
 Future value
 $= 5.4163 \times \$12\,000$
 $= \$64\,995.60$ (1 mark)
- ii Total contributions
 $= 5 \times \$12\,000$
 $= \$60\,000$
 Interest
 $= \$64\,995.60 - \$60\,000$
 $= \$4\,995.60$ (1 mark)

- 13 Number of quarters = $6 \times 4 = 24$
 Interest rate per quarter
 $= 3\% \div 4 = 0.75\%$ ✓
 Contribution required
 $= 0.0382 \times \$75\,000$
 $= \$2\,865$ ✓
 (2 marks)

- 14 i For 74 months at 0.0080 per month the interest rate factor is 55.684 46
 Present value
 $= \$200 \times 55.684\,46$
 $= \$11\,136.892$
 $= \$11\,137$ (nearest dollar)
 The present value of the annuity is \$11 137 to the nearest dollar. (1 mark)
- ii rate = 10.8% pa = 0.9% per month
 So $r = 0.0090$
 6 years
 $= 6 \times 12$ months
 $= 72$ months
 So $N = 72$
 The required interest factor is 52.821 18 ✓
 Amount of each repayment
 $= \$21\,500 \div 52.821\,18$
 $= \$407\,033\,693...$
 $= \$407$ (nearest dollar) ✓
 (2 marks)

- 15 Number of periods = $2 \times 2 = 4$
 Interest rate per period
 $= 4\% \div 2 = 2\%$
 Interest rate factor = 4.1216
 $FV = \$25\,000 \times 4.1216$
 $= \$103\,040$

Answer C

- 16 Using 3% per annum
 $= 1.5\%$ per half-year:
 Total amount
 $= 1200 \times 1.015^3 + 1200 \times 1.015^2$
 $+ 1200 \times 1.015 + 1200$
 $= 4909.08405...$
 $= 4909.08$
 $\therefore$ there is \$4909.08 in her account
 Answer A

- 17 Rate = 10% pa
 $= 5\%$ per six-months
 Period = 3 years
 $= 6$ six-months ✓
 Value of investment
 $= \$5000 \times 1.340$
 $= \$6700$ ✓
 (2 marks)

- 18 i Future value of \$1 = \$6.4684
 $\therefore$ Future value of \$5000
 $= \$6.4684 \times 5000$
 $= \$32\,342$ (1 mark)
- ii Future value of \$1 = \$8.1420
 Annuity which yields \$407 100
 $= \$407\,100 \div 8.1420$
 $= \$50\,000$ (1 mark)
- iii \$1000 invested at 4% per annum compounded quarterly for 2 years is equal to \$1000 invested for 8 periods at 1% per period. ✓
 $\therefore$ Future value
 $= 1000 \times \$8.2857$
 $= \$8285.70$ ✓
 $\therefore$ Interest
 $= \$8285.70 - 8 \times \1000
 $= \$285.70$ ✓
 (3 marks)

- 19 i \$1 invested at the end of every year for 4 years at 5% p.a. compounded annually becomes \$4.3101. ✓
 $\therefore$ Rod's investment will become $\$4.3101 \times 3600$
 $= \$15\,516.36$ ✓
 (2 marks)
- ii Interest
 $= \$15\,516.36 - 4 \times \3600 ✓
 $= \$15\,516.36 - \$14\,400$
 $= \$1116.36$ ✓
 (2 marks)